

Contrast Duality of Adversarial Learning in Network Intrusion: A Review

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Abstract—AI-based solutions are instrumental in cybersecurity, harnessing their ability to analyze vast datasets, identify complex patterns, and detect anomalies. However, attackers can exploit these capabilities to plan and execute sophisticated attacks, posing significant challenges to traditional security measures. Adversarial attacks pose a concerning threat to cybersecurity, especially when targeting machine learning model vulnerabilities. Adversarial learning exhibits a contrasting duality in network intrusion detection, serving as both an advanced offensive strategy for exploiting system weaknesses and a crucial defensive tool for identifying and mitigating adversarial threats. Our study presents an in-depth overview of focused adversarial learning threats such as data poisoning, test time evasion, and reverse engineering, and how adversarial training is applied as a defense mechanism. The convergence of adversarial learning attacks and defenses in network traffic data, alongside the advancements in machine learning and deep learning techniques, is still a relatively underexplored domain. Our research lays the groundwork for strengthening defense mechanisms to address the potential breaches in network security and privacy posed by adversarial learning attacks. Through our in-depth analysis, we identify domain-specific research gaps, such as the scarcity of real-life attack data and the evaluation of AI-based solutions for network traffic. Our focus on these challenges aims to stimulate future research efforts toward the development of resilient AI-based defense strategies.

Impact Statement—Understanding the evolving nature of network intrusion attacks and defenses is paramount for organizations to proactively protect their networks, respond effectively to security incidents, comply with regulations, foster innovation, and continuously improve their cybersecurity practices in an ever-changing threat landscape. Network intrusion attacks in the healthcare, finance, and national security sectors can result in compromised data privacy, financial losses, operational disruptions, threats to public safety, and national security risks. Adversarial learning attacks and defenses are pivotal in securing data and network operations by ensuring the reliability of outputs from AI-driven cybersecurity applications. This review offers a comprehensive analysis of adversarial learning attacks and

defenses, focusing on training, inference, and model analysis. It provides an understanding of the dynamic nature of these attacks, major threats and underscores the importance of ongoing research to strengthen network intrusion prevention and detection through robust AI solutions.

Index Terms—Adversarial learning, artificial intelligence in computer networks, artificial intelligence in cyber-security, deep learning, machine learning.

I. INTRODUCTION

CYBERSECURITY is critical in today's interconnected world, where cyber threats, such as malware, phishing, ransomware, and data breaches, continue to evolve and pose significant risks to individuals, organizations, and even national security [5], [88]. The integration of embedded systems has revolutionized critical sectors such as Healthcare, Automated Vehicles, and National Defense, relying on advanced technology and data integrity for improved efficiency, safety, and security. Nonetheless, the integration has also exposed network components more susceptible to cyber attacks and restricting their practical and secure implementation [82], [115]. Current trends indicate that approximately 67% of the world's population has access to the internet, with social media platforms playing a significant role in this accessibility [95]. However, this increased connectivity has also expanded the attack surface for adversaries, resulting in heightened cybersecurity threats. Notably, the economic impact of cyber attacks has grown substantially, with a global estimated loss of approximately \$6 trillion in 2021, doubling the costs recorded in 2015 [5]. Worldwide end-user spending on security and risk management is projected to total \$215 billion in 2024, an increase of 14.3% from 2023 [7].

The rapid progress in computational capabilities and the utilization of big data have positioned machine learning as a critical element in modern defense strategies. Nevertheless, the susceptibility of ML/DL-based solutions to adversarial attacks undermines current defense protocols. Developing adaptive defensive strategies that can effectively counter evolving threats remains a persistent challenge [49]. Through our evaluation, we identify the current research gaps in adversarial learning pertaining to network intrusion detection systems (NIDS). Our contribution lies in providing a baseline understanding of the existing research breadth and presenting the potential future directions in the field of ML/DL-based NIDS. The objective is to facilitate the development of robust ML/DL-based NIDS that effectively harness evolving technological capabilities, while

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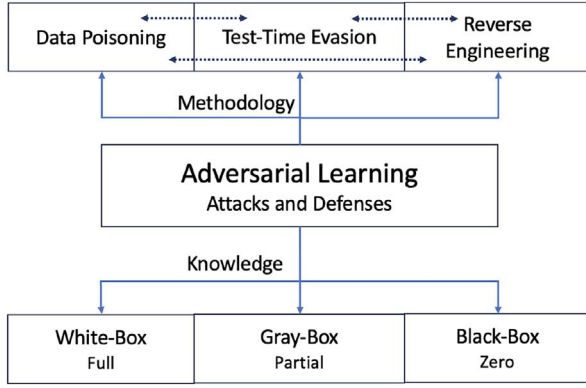


Fig. 1. Data poisoning, test-time evasion, and reverse engineering adversarial learning attacks and defenses for ML/DL-based NIDS.

also demonstrating resilience against both known and emerging security threats.

A. Research Objectives

Our work is driven by the growing need to understand the influence of artificial intelligence (AI) methodologies, specifically, adversarial learning, and technological advancements on innovative research and their practical implications for cybersecurity. To stay current and aligned with the evolving cybersecurity landscape and leverage emerging technologies, we aim to achieve the following research objectives.

- 1) Presenting a taxonomy of adversarial learning, NIDS, ML/DL techniques, attacker expertise, and NIDS datasets to utilize in efficient adversarial attack and defense strategies.
- 2) Reviewing the existing research on employing adversarial learning in data poisoning (DP), test-time evasion (TTE), and reverse engineering (RE) attacks in network intrusion domain.
- 3) Assessing available defense strategies against adversarial attacks in NIDS, emphasizing their limitations and potential future directions.

As presented in Fig. 1, we explore ML/DL-based adversarial attacks and defenses within the context of DP, test time evasion, and RE. Through analyzing these three key subdomains in adversarial learning, we cover a comprehensive exploration of adversarial attacks in machine learning from training data manipulation to inference-time deception and model vulnerability analysis. We examine the various threats, attacks, and possible defenses in which both attackers and defenders may possess different levels of knowledge, such as white-box (full), black-box (zero), and gray-box (partial) understanding of the model, algorithm, and training data. DP can be instrumental in training and test time evasion attacks. Similarly, RE can provide significant details to sharpen the attacks. At the same time, by analyzing the model's vulnerabilities and threats to data integrity, RE can be pivotal to develop robust defense mechanisms. Table I summarizes the primary categories of reviewed attacks, their corresponding stages in the ML pipeline, and their impacts on data and the model.

B. Our Contributions

Our work provides a focused and in-depth review of adversarial learning within the context of network intrusion detection systems (NIDS), specifically targeting three major subdomains: 1) DP; 2) TTE; and 3) model RE. We concentrate on the network intrusion domain, where the implications of adversarial attacks are particularly severe and underexplored. We explored the dual nature of adversarial learning, examining its application in both offensive strategies and defensive mechanisms under these subdomains of network intrusion domain.

We systematically analyze research conducted between 2018 and 2024, capturing both foundational breakthroughs and recent innovations. This timeframe allows us to review the latest developments in the field and track how it has progressed. By synthesizing findings across these three attack vectors, we identify persistent vulnerabilities, evaluate the robustness of existing defenses, and highlight emerging trends and gaps.

- 1) *Comprehensive Analysis:* We provide a detailed review of existing literature, offering valuable insights into adversarial learning-based attacks and defenses in NIDS.
- 2) *In-Depth Investigation:* We examine DP, TTE, and RE adversarial attacks on NIDS, identifying emerging trends and novel techniques in this domain.
- 3) *Highlighting Research Gaps:* Our study identifies critical gaps in the practical implementation of advanced ML/DL methodologies for NIDS, focusing on real-world challenges.
- 4) *Foundation for Future Research:* We synthesize the current state of research and lay the groundwork for developing innovative, resilient defense mechanisms for real-world NIDS deployments.

C. How Our Work Is Different?

To the best of our knowledge, there is no existing exhaustive examination of adversarial learning attacks and defense strategies within the network intrusion domain, complete with a detailed exploration of DP, TTE, and RE. To address this critical gap, we present the threats to various stages of machine learning pipeline and potential defense strategies. Table II serves as a quick reference to illustrate the scope of our work and highlights the extent of its comprehensiveness compared with the reviewed literature. We not only cover the necessary background on adversarial learning and NIDS for a domain specific review, but keep it broad enough to cover the various machine learning pipeline threats (DP, test time evasion, and RE), and how those can have different severity impact with the attacker's knowledge (black/white/gray box).

It is important to note that the existing literature predominantly concentrates on adversarial attacks and defenses in image, audio, and video domains. In contrast, research specifically dedicated to adversarial learning in the context of NIDS is limited, accounting for less than 10% of all adversarial learning research.

TABLE I
REVIEWED ADVERSARIAL ATTACKS

Attack Category	ML Pipeline Stage	Target	Primary Goal
Data poisoning	Training phase	Training data and model	Compromise the model's learning process
Test-time evasion	Inference phase	Model predictions (output)	Misclassify during inference
Reverse engineering	Any phase	Model parameters and behavior	Extract insights to facilitate further attacks

TABLE II
COMPARISON OF RECENT SURVEYS IN ADVERSARIAL LEARNING AND ADVERSARIAL LEARNING IN NIDS

Reference	ML/DL in NIDS	Adversarial Learning in NIDS	NIDS Taxonomy	Adversarial Attacks	Adversarial Defenses	NIDS Datasets	Data Poisoning	Test-Time Evasion	Reverse Engineering	Black Box/ White Box/ Grey Box	Challenges in NIDS Adversarial Learning
Zhang et al., 2020 [118]	None	None	None	High	Low	None	None	None	None	B/W	None
Aldweesh et al., 2020 [12]	Medium	None	Medium	High	None	Medium	None	None	None	None	High
Miller et al., 2020 [64]	None	None	None	High	High	None	High	High	High	B/W/G	Medium
Han et al., 2021 [40]	High	High	None	High	High	Low	None	High	None	B/G	Low
Alatwi and Morisset, 2021 [11]	Low	High	None	Medium	Medium	Medium	High	High	Low	B/W/G	High
Chakraborty et al., 2021 [30]	Medium	Low	High	High	High	None	High	High	Low	B/W/G	Low
Zhou et al., 2022 [120]	None	None	None	High	High	None	Low	Low	None	B/W	None
Jmila and Khedher, 2022 [48]	High	Medium	Medium	High	High	Low	Low	Low	None	B/W/G	High
McCarthy et al., 2022 [59]	High	High	Low	High	High	Low	None	High	None	B/W/G	High
Apruzzese et al., 2022 [18]	High	High	None	High	None	None	High	High	High	B/W/G	None
He et al., 2023	High	High	High	High	High	Low	None	High	None	B/W	High
Vitorino et al., 2023 [105]	High	High	None	High	High	None	High	High	High	B/W/G	High
Ibitoye et al., 2023 [45]	High	High	Medium	High	High	None	High	High	High	B/W/G	Low
Alotaibi and Rassam, 2023 [13]	Medium	High	Medium	High	High	Medium	High	High	None	B/W/G	High
Our work	High	High	High	High	High	High	High	High	High	B/W/G	High

D. Structural Overview

The subsequent sections of this article are structured as follows.

Section II reviews related work in the field, discussing prior contributions and identifying how our study distinguishes itself. Section III provides a comprehensive background on network intrusion, including a taxonomy of adversarial learning, crafting adversarial samples, and utilizing in adversarial attacks and defenses in NIDS. It also provides the attack categories, attacker's knowledge and an overview of subdomains we cover, establishing the foundation for the subsequent analysis. Section IV outlines the research methodology, explaining the rationale behind the selection and inclusion of the works we review. Section V presents our core review, analysis, and results of adversarial learning attacks and defenses, focusing on DP, TTE, and RE. Section VI discusses the limitations and challenges encountered within the scope of our review. Section VII offers a broader discussion, summarizing key insights, and Section VIII presents the conclusion with the proposed future directions.

II. RELATED WORK

Initial adversarial learning vulnerabilities were primarily studied within the computer vision domain. However, it is crucial to recognize domain-specific nuances when assessing potential attacks and defenses in cybersecurity. Table II pro-

vides a summary of related works, evaluating their coverage of adversarial learning, adversarial attacks, and countermeasures, and assessing their alignment with network intrusion-specific knowledge. We review the related work aligned with our research scope and discuss how our contribution stands out in the field.

Zhang et al. explored natural language processing (NLP) targeted adversarial attacks, and compared algorithms and models across various domains [118]. Rosenberg et al. highlighted the divergence between adversarial learning attacks and defenses in image-based systems and those specific to cybersecurity, emphasizing the distinct nature of data features involved [80]. Unique challenges in network intrusion detection include managing high volumes of real-time data, minimizing false positives and false negatives, inspecting encrypted communications, adapting to evolving attack techniques, mitigating adversarial manipulation of detection models, ensuring scalability in large networks, and detecting insider threats. While new defense methods are continually evolving to address these challenges, the need for NIDS to swiftly and accurately detect specific threats in real time remains a significant challenge. There is a limited research addressing these challenges, including adversarial learning vulnerabilities, exploitation opportunities in ML/DL models, feasible threat surfaces and attack vectors, and countermeasures against adversarial attacks in network intrusion domain.

There are numerous directions and focuses in the field of adversarial learning attacks and defenses with a goal to compiling the knowledge to develop robust solutions and proposing potential directions to strengthen domain knowledge. For instance, Zhou et al. explored adversarial attacks and defenses in deep learning within the cybersecurity domain, focusing on Advanced Persistent Threats (APT). Their research centered on constructing a framework detailing a five-stage APT lifecycle for AL attacks and defenses. While they applied this framework to domains such as image, video, audio, and text, the exclusion of network traffic data discussions limits its applicability to network intrusion [120]. In another recent survey on NIDS adversarial attacks, He et al. emphasized critical issues arising from outdated benchmark datasets and the difficulty in generating authentic network features due to data instability. Moreover, the impracticality of modifying features for attacks lies in their inability to be transferable. They also discussed the scarcity of defense mechanisms against packet-level attacks [41].

Jmila and Khedher studied shallow ML classifiers' efficacy in assessing robustness, enhancing them with Gaussian data augmentation. Their research suggested tailoring methods, parameters, and datasets specific to network intrusion detection scenarios, noting performance degradation with certain datasets and attack types [48]. A survey by McCarthy et al. explored functionality-preserving adversarial attacks and defenses in the area of poisoning, evasion and transferability attacks in cybersecurity domain [59]. Vitorino et al. summarized the state-of-the-art approaches focusing on adversarial learning strategies for generating realistic adversarial examples to analyze and protect real-life scenarios. However, their work identified that more than 75% of the reviewed research used known common methods and lacked novel strategies to generate adversarial samples [105].

Ibitoye et al. categorized NIDS attacks into problem and feature space, with feature space representing data variables and problem space denoting input types such as files or images. The study also mapped specific ML methods, such as supervised learning, to distinct ML task categories such as classification [45]. In a study, Apruzzese et al. revealed the limitations of utilizing deep learning methods for practical NIDS attacks. They discussed that feature-space attacks for NIDS are impractical in practice due to the difficulty of deriving adversarial traffic from features [18]. Alotaibi and Rassam examined adversarial machine learning attacks and defense strategies by analyzing individual research works, presenting each strategy as outlined in specific studies rather than categorizing them into broad groups [13].

III. BACKGROUND AND TAXONOMY

A. Adversarial Learning

Approximately a decade ago, researchers identified adversarial learning as a significant vulnerability in neural networks. Adversarial learning pertains to their susceptibility to small, imperceptible changes in input data, as well as the manipulation of decision boundaries, which can lead to misclassifications [20], [39], [97]. In past few years, adversarial learning has grown as a field of great research interest because

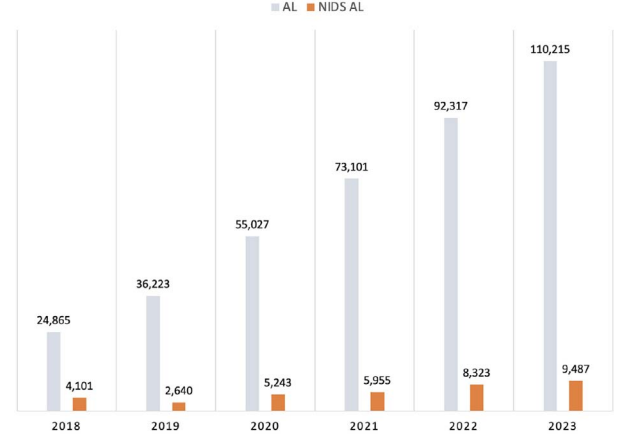


Fig. 2. 2018–2023: Adversarial learning research trends [84].

of its wide applicability and impact on artificial intelligence-based solutions in different domains. As depicted in Fig. 2, the field of adversarial learning has experienced a significant five-fold growth in the past five years. However, research in adversarial learning in network intrusion is still limited [84]. Search results may exhibit bias based on keyword selection and the emphasis on titles and abstracts. Nonetheless, they offer a foundational insight into trends within the research area.

B. Crafting Adversarial Samples and Their Contrast Utilization in Attack and Defense

Adversarial examples are crafted inputs designed to deceive machine learning models by introducing subtle, imperceptible perturbations that lead to incorrect predictions. Given a model $f(x)$ that predicts a label y for an input x , an adversarial attack can be formulated as follows.

Objective: Generate an adversarial example x' such that

$$x' = x + \delta$$

where δ is a small perturbation added to the original input x .

Fast Gradient Sign Method (FGSM): FGSM perturbs the input x in the direction of the gradient of the model's loss function $J(x, y; \theta)$, calculated with respect to x . The method is defined as

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x J(x, y; \theta))$$

where ϵ is the perturbation magnitude, $\nabla_x J$ is the gradient of the loss with respect to x , and $\text{sign}(\cdot)$ represents the gradient's direction.

Generative Adversarial Networks (GANs): A GAN comprises a generator G and a discriminator D . The generator G creates synthetic data x' , aiming to make it indistinguishable from real data x , while the discriminator D seeks to differentiate between real and generated data. The GAN's objective is

$$\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

where $p_{\text{data}}(x)$ represents the real data distribution, z is a noise vector sampled from $p_z(z)$, $G(z)$ generates synthetic data, and $D(x)$ outputs the probability that x is real.

In adversarial learning, the generator G is trained to produce inputs x' that maximize the misclassification rate of a target model f . By optimizing x' to exploit the vulnerabilities of f .

Transferability and Immobility: The transferability of an adversarial perturbation enables it to deceive multiple different models. The immobility reflects a model's unique strength, as its specific robustness prevents attacks from transferring from other models.

C. NIDS ML Pipeline Vulnerabilities

DP: An attacker injects malicious or misleading data into a machine learning model's training data to corrupt model's behavior. Adversarial data is injected through flipping target class labels (label-flipping attack), manipulating features but keeping the class label unchanged (clean-label attack), inserting backdoors, or flooding with noisy data to evade detection of specific attacks or to degrade the system's overall accuracy.

TTE: An attacker creates malevolent network traffic that circumvents detection during inference, all without changing the training data. These attacks capitalize on vulnerabilities in the trained model by subtly adjusting input features such as packet size, timing, or payload patterns to influence the model's actions and produce inaccurate outputs.

RE: An attacker reconstructs a copy of the model by querying it or reconstructing input features from the outputs. Surrogate model without authorization can lead to the theft of intellectual property. Membership Inference Attacks can expose sensitive information in the training data violating the data privacy.

We analyze all three NIDS subdomains as DP affects the training phase, TTE targets the inference phase, and RE primarily impacts inference but can also reveal vulnerabilities from the training phase.

D. Adversarial Learning Attacks

Adversarial attacks fall into different categories based on methodologies and approaches, which directly influence the strategies employed by attackers to create adversarial examples and exploit vulnerabilities [117], [120]. Targeted attacks aim to evade detection for specific malicious activities, while untargeted attacks disrupt the system by triggering false alarms or misclassifying normal traffic as malicious [29], [120].

- 1) *Untargeted Attacks:* Cause misclassification ($f(x') \neq f(x)$).
- 2) *Targeted Attacks:* Force the model to predict a specific target class y_t ($f(x') = y_t$).

Attacker's knowledge of model and the training data categorizes attacks as black-box (no knowledge), white-box (full knowledge), and gray-box (partial knowledge) attacks. Fig. 3 illustrates the categories in relation to the attacker's knowledge, attack surface, and impact severity.

Black-box: A black-box approach assumes that the adversary has no access to the internal parameters or gradients of the

target model or training data. Instead, the attacker can only query the model with inputs x and observe the outputs $f(x)$, where f is the unknown decision function. In the context to adversarial learning, it can be challenging to craft efficient adversarial samples without the knowledge of gradients. Black-box approach can also be computationally expensive to make a large number of queries to gain some meaningful insight.

White-Box: A white-box approach assumes that the adversary has complete knowledge of the target model, including its architecture, algorithms, internal parameters, and access to the training data. For example, the attacker can observe and utilize the function $f(x; \theta)$, where θ denotes the model parameters, and may also exploit the training data distribution $\mathcal{D}_{\text{train}}$ to manipulate adversarial learning.

Gray-Box: In a gray-box approach, the adversary has partial knowledge of the target model. This typically includes access to the training data distribution $\mathcal{D}_{\text{train}}$ or some information about the model architecture or hyperparameters, but not the exact internal parameters θ of the deployed model. The attacker may exploit this limited knowledge to train a surrogate model $\hat{f}(x; \hat{\theta})$ that approximates the behavior of the target model $f(x; \theta)$.

Model manipulation attacks involve exploiting the transferability of adversarial attacks between models, corrupting the model, deceiving model through feedback mechanisms, and introducing malicious patterns during dynamic updates [43], [50]. Other major classifications are based on network traffic manipulation strategies: feature-based, flow-based, and packet-based [81].

Hybrid adversarial attacks on deep learning models involve combining multiple attack techniques to compromise model integrity and performance. These attacks can include a combination of evasion and poisoning techniques, as well as targeting strategies [37]. Adaptive adversarial attacks continuously adapt to countermeasure updates and target NIDS feedback timing to compromise the system's learning process and effectiveness [110]. Adversarial NIDS attacks include strategic manipulation of network traffic, subtle perturbations to evade detection algorithms, generative attacks that generate malicious traffic resembling benign patterns, and stealth attacks that hide malicious activities within legitimate traffic to avoid detection. [11], [91], [96], [103], [111].

E. Adversarial Learning Defenses

Adversarial learning defenses have advanced over the years, encompassing techniques and strategies to fortify machine learning models against attacks. Approaches such as adversarial training, detection/filtering, gradient masking, and randomized smoothing can provide effective defenses. Training-based defenses enhance adversarial robustness through adversarial training, robust feature extraction, model architecture modifications, compression and quantization, and diverse loss functions. These techniques enhance models against attacks by improving training data, resilient feature extraction, network architecture adjustments, model simplification, and optimization diversity

Attacker's Knowledge Attack Surface Attack's Severity	Black Box	Gray Box	White Box
	- No access to Model's structure or the algorithm applied. - No access to training data - Query-based probing	- Partial knowledge of Model's structure or the algorithm applied. - Partial knowledge of the training data - Query-based probing - Potential Model/algorithm manipulation - Potential Data manipulation	- Full access to Model's structure and/or the algorithm applied. - Full access to training data - Manipulation of Model's structure, parameters, and algorithms applied - Manipulation of training and testing data
	- Data Poisoning: Possible through iterative probing and observation. - Test Time Evasion: Viable by crafting adversarial samples and observing model responses to refine attacks. - Reverse Engineering: Through probing the model with crafted queries to gain insights into its behavior.	- Data Poisoning: Allowing attackers to craft poisoned data strategically. - Test Time Evasion: Attacks can leverage partial model knowledge to craft inputs. - Reverse Engineering: With some knowledge attackers may attempt reverse engineering to uncover vulnerabilities or manipulate the model's behavior.	- Data Poisoning at training - Test Time Evasion at inference - Reverse Engineering by analyzing the model's internals, attackers can identify vulnerabilities and devise effective attacks.
	Low- Medium	Medium- High	High- Critical

Fig. 3. Attacker's knowledge, attack surface and attack's severity in case of black-box, gray-box, and white-box scenarios.

[23], [119]. Input transformation and reconstruction-based defenses enhance adversarial robustness by altering and reconstructing input data, guarding against perturbations and boosting model generalization. Ensemble-based defenses strengthen robustness by using multiple models, diversifying ensemble members through varied training to collectively improve defense efficacy against attacks [102].

Randomized Smoothing introduces random noise to input data during training and inference to enhance the model's decision boundaries and reduce vulnerability to adversarial perturbations [34]. Example distributions to differentiate and eliminate adversarial samples are another defense mechanism [114].

Targeted adversarial defenses are more efficient for domain-specific scenarios. However, more generalized approach can expand the applicability of these defense strategies.

F. Machine Learning in Network Intrusion Detection

Since the late 1990s and early 2000s, early applications of machine learning focused on network intrusion detection and malware detection through file analysis and behavioral patterns. Over time, advanced machine learning techniques have found extensive applications in anomaly detection, threat intelligence, fraud detection, and spam filtering. Deep learning, a subset of machine learning, leverages neural networks to autonomously extract features and patterns from raw data, enhancing cybersecurity capabilities [24], [93]. Signature-based detection and anomaly-based detection are two broad categories of network intrusion detection methods [11], [47]. Signature-based network intrusion detection systems (SNIDS) detect intrusions by matching network traffic against preinstalled attack signatures. They are effective in detecting known attacks but struggle with unknown or new attacks [52]. Anomaly-based network intrusion detection systems (ADNIDS) detect intrusions by identifying deviations from normal network patterns. They are effective in detecting unknown or new attacks but may have higher false-positive rates [47], [67].

Different machines and deep learning models can be valuable for NIDS, but their usefulness depends on meeting specific security goals and having the necessary resources. Convolutional neural networks (CNNs) enable the learning of hierarchical representations of network traffic patterns, improving the ability to identify anomalies [19], [58]. Deep neural networks (DNNs) are more generalized and can process various types of data, including numerical, categorical, or textual data [56], [58], [71], [106], [108]. Recurrent neural networks (RNNs) are suitable for processing sequential data, making them applicable in NIDS for analyzing network traffic over time and detecting anomalies based on deviations from learned temporal patterns [16].

Autoencoders and variational autoencoders (VAEs) are unsupervised deep learning models used in NIDS to learn efficient representations of normal network traffic. By training autoencoders on normal traffic, deviations or intrusions can be detected by measuring the reconstruction error [53], [65]. Graph neural networks (GNNs) hold promise for network intrusion detection due to their ability to handle complex dependencies and relationships in network data [107]. Specialized deep learning architectures for NIDS that combine multiple layers to capture complex patterns and relationships in network traffic data, can enhance intrusions and anomaly detection capabilities [116].

Transfer learning techniques have been applied in NIDS by utilizing pretrained deep learning models. These models, originally trained on large-scale datasets such as ImageNet, are fine-tuned on network traffic data to leverage the learned representations [32]. Federated Learning in NIDS revolutionizes network intrusion detection by enabling decentralized devices to collaboratively train models. This approach enhances detection accuracy and robustness without compromising data privacy or requiring data centralization [8].

G. Network Intrusion Benchmark Datasets

Ensuring the relevance of benchmark datasets is vital for adapting to evolving threat scenarios. Algorithms may not yield

TABLE III
NETWORK INTRUSION DATASETS, RELATED WORKS, AND BRIEF DETAILS

Dataset	Related Works	Brief Details
KDD99	[15], [38], [99], [100]	Early, widely used dataset; suffers from redundancy (75%) and outdated attack scenarios.
NSL-KDD	[77], [89]	Improved KDD99 by removing redundancy; simulated dataset, lacks real-world applicability.
Kyoto 2006+	[38], [94]	Real-world traffic dataset; includes normal, known, and unknown attacks using honeypots and sensors.
ISCX 2012	[38], [75], [92], [100]	Real-world flow-based dataset covering DoS, DDoS, Brute-force, and Infiltration attacks.
UNSW-NB15	[2], [66]	Modern dataset blending real activities and synthetic attacks; nine attack types such as Fuzzers, DoS, and Exploits.
CIDDS-001	[78]	Virtual cloud-based dataset; features DoS, Port Scan, Ping Scan, and Brute-force attacks.
CICIDS2017	[89]	Flow-based dataset with normal and attack instances (e.g., DoS, DDoS, Brute-force, Port Scan).
CSE-CIC-IDS2018	[3], [89]	Covers seven attack types, including Brute-force, Botnet, and Web attacks, with modern scenarios.
CICDDoS2019	[90]	Focused on DDoS attacks with realistic background traffic; valuable for modern DDoS research.
CIC IoT 2023	[69]	Tailored for IoT; includes attack types such as DDoS, Recon, Spoofing, and Mirai.

desired results if the data lacks reliability or fails to accurately represent the threats, impeding the development of robust models against adversarial attacks. Several network intrusion benchmark datasets have been developed over the years to cater to the diverse and evolving requirements of network intrusion detection and prevention strategies. Table III presents a summary of the major datasets commonly utilized in reviewed research works. The KDD99 dataset, one of the oldest and widely utilized datasets in NIDS research, represents four fundamental attack scenarios of DoS, Probe, R2L, and U2R. While it played a crucial role in early research, the dataset suffers from highly redundant data (above 75%) and outdated attack scenarios. [15], [38], [99], [100]. NSL-KDD is essentially KDD99 dataset without redundant data. Both datasets, KDD99 and NSL-KDD, are simulated virtual network datasets and do not accurately represent real-world data. Consequently, they are not reliable for reflecting modern real-world attacking scenarios [77], [89]. Kyoto 2006 dataset was built capturing real-world network traffic as not-labeled data. Using honeypots, darknet sensors, email server, web crawler and network security mechanisms to detect attempts of unauthorized use, Kyoto 2006 dataset includes normal, known and unknown attacking instances [38], [94].

ISCX 2012 is a real-world, flow-based labeled dataset to represent actual network attack scenarios in four major categories of DoS, DDoS, Brute-force, and Infiltration [38], [75], [92], [100]. UNSW-NB15 dataset was created using the IXIA PerfectStorm tool at UNSW to simulate a mix of real modern normal activities and synthetic attack scenarios. The dataset consists of nine types of attacks, including Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms [2], [66]. CIDDS-001 dataset (2017) was developed in a virtual cloud-based environment by executing DoS, Port Scan, Ping Scan, and Brute force attacks [78]. CICIDS2017 dataset is also labeled flow-based dataset with normal and attack instances representing DoS, DDoS, Brute-force, Port Scan, Bot, Web Attack, and Infiltration attacks [89]. CSE-CIC-IDS2018 consists of seven attacking scenarios of Brute-force, Heartbleed, Botnet, DoS, DDoS, Web Attack, and Infiltration of the network from inside [3], [89]. CICDDoS2019 is a labeled flow-based dataset, specific to more recent DDoS

attacks with emphasizing on generating realistic background traffic [90]. CICIoT2023 dataset presents seven distinct attacks, namely DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, and Mirai [69].

IV. RESEARCH METHODOLOGY

To conduct a comprehensive search for research publications across multiple research repositories, we employed a strategic approach that involves the utilization of multiple keywords and phrases. By carefully selecting and combining relevant terms, we aim to cast a wide net and ensure the inclusion of diverse and pertinent literature. Our search strategy involves identifying key concepts and themes related to our research topic and constructing queries that encompass these facets. To ensure a thorough analysis, we consulted renowned research databases, including *IEEE Xplore*, *ACM Digital Library*, and *SpringerLink*. We also utilized Google Scholar for the existing work in the field, additionally querying *Base* and *arXiv*. In addition to these databases, we leveraged *Dimensions.ai*¹ for its comprehensive research insights, allowing us to gather a broader perspective on the relevant literature [6]. Fig. 4 shows a step by step to identify, filtering, screening, retrieving, and selecting the studies specific to the scope of the article.

Our base search was built around *adversarial learning attacks and defenses in network intrusion detection systems*, and then we narrowed it down towards *DP*, *TTE*, and *RE*. We further categorized these topics into *black-box*, *white-box*, and *gray-box* adversarial learning attacks and defenses in NIDS. To achieve an extensive coverage of relevant publications in adversarial machine learning, a combination of the adversarial learning keyword with other appropriate terms such as perturbation, evasion, inference, model inversion, model stealing, and model poisoning was employed. Additionally, concepts closely tied to network intrusion detection, including anomaly detection, cyberattack, and wireless and IoT networks, were taken into account to ensure a comprehensive scope. We executed around hundred initial AL-ML-DL-NIDS queries, and then we iterated and refined our search queries, incorporating synonyms, related terms, and variations to capture a

¹<https://app.dimensions.ai/discover/publication>

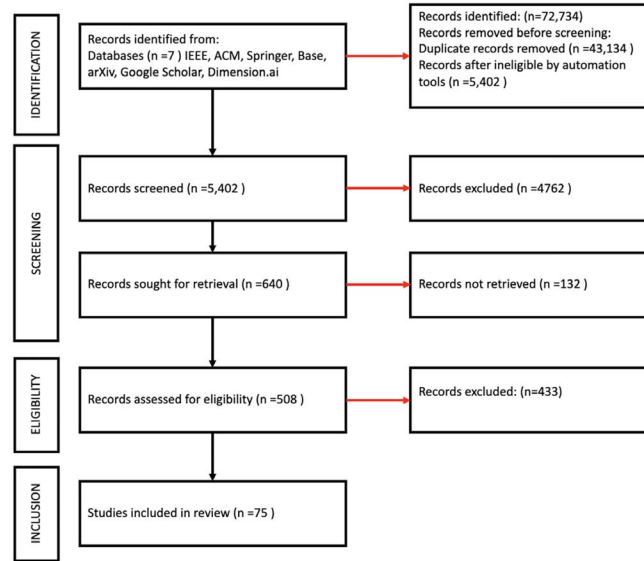


Fig. 4. PRISMA framework for identification, screening, eligibility, and inclusion/exclusion of the studies.

TABLE IV
RESEARCH LITERATURE- INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria	Key Considerations
Relevance to adversarial learning in NIDS	Papers directly address adversarial learning techniques in the context of network intrusion detection systems
Focus on adversarial attacks and defenses	Papers discuss adversarial attacks (e.g., data poisoning, test-time evasion, reverse engineering) and defenses specific to NIDS
Methodological alignment	Papers focusing on machine learning, deep learning, and AI methodologies in the context of NIDS
Specific attack types	Papers investigating data poisoning, backdoor attacks, test-time evasion, and reverse engineering attacks in NIDS
Diversity of defense strategies	Papers exploring various defense mechanisms against adversarial attacks in NIDS
Attacker's knowledge and their impact	Papers discussing black box, white box, and gray box adversarial attacks and defenses in NIDS
Dataset utilization	Papers utilizing NIDS datasets or proposing new datasets for research purposes
Exclusion Criteria	
Irrelevant topics	Papers not directly related to adversarial learning in NIDS
General machine learning or deep learning studies	Papers that do not specifically focus on NIDS or adversarial learning in the context of network security
Nontechnical papers	Non-technical papers, such as opinion pieces, editorials, or reviews without novel research contributions
Outdated material	Papers published before a certain date (e.g., more than 5 years old) to focus on the most recent research
Language and publication quality	Poorly written or low-quality papers, as well as papers not published in reputable journals or conferences

broader range of relevant publications. Through this meticulous process of applying inclusion and exclusion criteria as per Table IV, we optimized the retrieval and compilation of research publications for this work, enabling us to gather a comprehensive and diverse collection of relevant studies from various sources.

V. RESULTS: ADVERSARIAL ATTACKS AND DEFENSES IN NIDS

Our work encompasses a comprehensive analysis of adversarial attacks and defenses within NIDS, focusing specifically on three critical attack vectors: 1) DP; 2) TTE; and 3) RE. We further categorize these attacks based on the adversary's level of knowledge, white-box, black-box, and gray-box, providing a detailed exploration of their methodologies, impacts, and corresponding defense strategies. The following sections discuss each of these attack categories, emphasizing their relevance to modern NIDS frameworks.

A. DP Attacks and Defenses

Data manipulation attacks pose a significant threat to deep learning neural network classifiers by undermining their integrity and detection capabilities [87], [104]. Adversaries utilize various techniques in these attacks, including DP, where malicious data is injected to corrupt the NIDS's training process. Additionally, adversaries manipulate the training data and process itself during the learning phase, weakening the NIDS's ability to accurately identify threats. Furthermore, attacks on feature extraction distort the representations of network data, making it harder for NIDS to identify malicious activities [63], [98].

DP attacks can manipulate network data by targeting port numbers, protocol types, or payload content to disrupt operations or deceive detection systems. For instance, Feature-based poisoning alters attributes to evade detection or induce false positives/negatives [104]. Packet-based poisoning alters

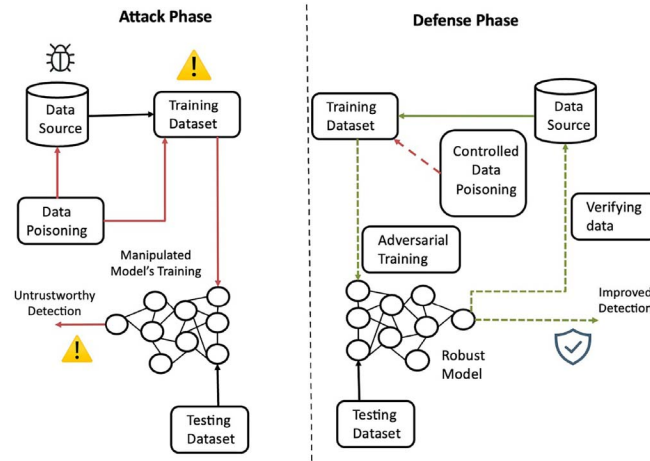


Fig. 5. Data poisoning adversarial learning attacks and defenses for ML/DL-based network intrusion detection systems.

individual packets, while flow-based poisoning manipulates traffic flows. Attackers can disrupt flows, introduce delays, and exploit flow-based protocol vulnerabilities. Content-based poisoning inserts malicious payloads. In techniques such as DNS poisoning, manipulation of DNS data occurs to deceive queries, steering users towards malicious websites through the false association of domain names with incorrect IP addresses. In Fig. 5, we illustrate how adversarial learning can be utilized both as a DP attack and a defense against DP attacks.

Backdoor DP Attacks: A backdoor attack can inject trigger patterns or perturbations into the training data or modify the model's architecture or parameters to manipulate the DL-based NIDS, causing it to exhibit malicious or unexpected behavior. Backdoor DP causes a model to misclassify test-time samples that contain a trigger. In this threat model, the attacker exerts control over the data by introducing poisons during the training phase and strategically inserting triggers during the inference phase [86].

Black-Box DP Attacks: A study by Kuppa et al. showed that by running manifold approximation on samples collected at attacker end for query reduction and understanding various thresholds set by underlying anomaly detector, an attacker can use spherical adversarial subspaces to generate attack samples. This black-box attack methodology is particularly effective when targeting anomaly detection systems that lack clearly defined decision boundaries between normal and abnormal classes, relying on a set of thresholds on anomaly scores to guide the decision-making process [53].

Ning et al. demonstrated that a stealthy backdoor black-box DP attack could be implemented with as low as 0.5% of the training data. In the loss-free digital attack scenario, this achieves an average attack success rate of over 91.1%. In physical attacks using lossy images, a trigger as small as 1% of the original image activates the backdoor with a success rate of over 78.5% under a 0.5% poison ratio [70]. A study by Li et al. achieved a success rate of 93.5% with their backdoor DP and RE attack on 54 deep learning image-based mobile apps. The attack incurred only a modest latency overhead of less than 2 ms

and resulted in a maximum accuracy decrease of 1.4% [56]. Xu et al. utilized Neural Machine Translation (NMT) demonstrating the feasibility of successful targeted backdoor DP attacks on black-box NMT systems with low poisoning rates of 0.006% for the language translation datasets [113].

White Box DP Attacks: Venkatesan et al. conducted a study on white-box flow-based poisoning availability attacks targeting a network scanning classifier. Using synthetic network data from the CyberVan testbed, they demonstrated that placing poisoned samples near 10% of high-confidence points with 20% DP reduced the model accuracy from 95% to below 50% [104]. A label flipping attack aims to identify a subset of N examples, where flipping their labels maximizes a specific objective function chosen by the attacker. In a white-box attack scenario, with the complete knowledge of the algorithm and data impact, Paudice et al. showed that classification error can be up to six times higher with 20% of DP through label flipping [73].

Wang et al. noted that a simple modification of the cross-entropy loss yields stronger poisoning attacks when using projected gradient ascent [109]. Alrawashdeh et al. demonstrated that an attacker can generate stealthy white-box adversarial samples using L-BFGS and FGSM Methods, and can also inject the trigger for backdoor DP attacks to degrade the model's accuracy. For instance, the researchers analyze the dataset to identify the triggers with the highest predictive power for selecting the correct labels. By manipulating these labels, attackers can effectively boost the success rate of their malicious activities [14].

Gray Box DP Attacks: Tolpegin et al. investigated targeted DP attacks in federated learning (FL) systems, where malicious participants (Insider Attacker) aim to poison the global model with mislabeled data. The attack is agnostic to the specific DNN architecture, loss function, or optimization function employed. It necessitates corrupting the training data, while the learning algorithm itself remains unchanged in this gray-box attack. The study reveals significant drops in accuracy and recall, even with a small number of malicious participants [101].

He et al. proposed a gray-box attack, *Liuer Mihou*, to use a surrogate deep learning model to modify the packet delay and injecting random packets in *Kitsune network attack dataset* and their own IoT dataset to generate stealthy adversarial samples by iterative operations to minimize the anomaly score to stay undetected by anomaly-based NIDS such as *Kitsune* [4], [42], [65]. An example of a gray-box backdoor DP attack was presented by Shafahi et al., termed as *Clean-label Attack*. In this scenario, the attacker possesses information about the model and its parameters but lacks insight into the training data. Through the manipulation of feature collisions, attackers can influence the classification process, resulting in a backdoor effect where the target class is erroneously classified as the base class [87]. Another study by Severi et al. showcased a backdoor attack where an attacker, armed solely with knowledge of the feature space, could launch a potent attack by adding a small set of poisoned samples, constituting just 1% of the training data [86].

DP Defenses: To defend against DP attacks in deep learning-based cybersecurity solutions, there are several known strategies to mitigate the risks. For example, data sanitization techniques preprocess and filter input data to remove or mitigate the effect of poisoned samples, while adversarial training exposes the model to adversarial examples during training to enhance its robustness. Model verification ensures the integrity of the trained model, ensemble learning combines predictions from multiple models for increased resilience, input validation detects, and filters out suspicious samples. Additionally, robust optimization techniques promote the learning of generalized and robust features.

Chen et al. introduced *De-Pois* that trained a mimic model using GANs to imitate the behavior of a target model trained on clean samples. By comparing the predictions of the mimic model and the target model, *De-Pois* could detect poisoned samples without prior knowledge of the machine learning algorithms or poisoning attack types [31]. A defense strategy proposed by Alrawashdeh et al. targeted white-box DP and backdoor DP attacks through activation function and neuron pruning, that could reduce the initial average loss of accuracy around 80% (10% to 2%) for Deep Belief Network (DBN) and around 85% (14% to 2%) for generative adversarial network (CoGAN) for NSL-KDD and ransomware datasets [14]. Another defense approach called *Nested Training* for NIDS was introduced by Venkatesan et al., using a diversified ensemble of classifiers trained on different subsets of the data. By leveraging disagreement among classifier predictions, the approach effectively mitigates DP attacks, with up to 30% of the training data being poisoned [104].

Schwarzschild et al. observed that models trained with Stochastic gradient descent (SGD) are significantly harder to poison, rendering poisoning attacks less effective in practical settings [85]. A defense strategy for label-flipping attacks, Paudice et al. proposed relabeling of suspicious points that may be indicative of malicious behavior [73]. Investigating the robustness of SGD against various DP attacks, Wang et al. demonstrated that SGD maintains optimal convergence rates on excess

risk even in the presence of DP [109]. Introducing the DPA-FL system, a two-phase defense mechanism for intrusion detection in federated learning, Lai et al. initially employed relative weight differences to compare participants' models, unveiling unique patterns that differentiate attackers from benign participants. Subsequently, the aggregated model was tested with the dataset to pinpoint attackers when accuracy dropped. DPA-FL identified and removed attackers within twelve rounds, even with a limited number of malicious actors. Study demonstrated that DPA-FL achieved 96.5% accuracy in defending against poisoning attacks and improved F1-score by 20% to 64% under backdoor attacks [54].

Table V represents the reviewed adversarial DP literature for various DP attacks and defenses employed against the known and potential adversarial attacks. Label-flipping poisoning is most common adversarial attack while detecting adversarial samples and filtering is one of the common defense strategies. Recent research trends indicate a growing preference among scholars for employing deep learning approaches over conventional machine learning methods. Regarding the suitability of the utilized IDS datasets, our analysis revealed that CIC-IDS2017 has garnered significant attention across multiple studies. This dataset, encompassing 80 network flow features and incorporating prevalent attack types such as Web-based, Brute force, DoS, DDoS, Infiltration, Heartbleed, Bot, and Scan, stands out as a contemporary and comprehensive resource for intrusion detection research [1], [89].

B. Test Time Evasion Attacks and Defenses

TTE attacks aim to deceive or manipulate a model's behavior during the inference phase, often by crafting adversarial examples or exploiting vulnerabilities in the model's decision-making process. The goal of the adversary is to intentionally deceive, bypass, or undermine the detection and defense mechanisms [25], [26], [39], [97].

Decision boundary manipulation is a primary TTE attack, where attackers deliberately manipulate the decision boundary of a machine learning model during testing. Attackers strategically modify input data to shift or distort the decision boundary, leading the model to make incorrect predictions. The model can be manipulated into making erroneous predictions (untargeted) or a specific misclassification (targeted), potentially causing security and reliability concerns. Adversarial inputs at test-time can exploit vulnerabilities in the detection algorithms or models to evade detection or mislead the system into classifying malicious activities as benign.

Fig. 6 demonstrates the fundamental approach of TTE attacks and how adversarial learning can be utilized as a defense mechanism against TTE attacks.

Black Box TTE Attacks: Aiken et al. addressed ML-based NIDSs deployed in Software-defined networks (SDNs), highlighting their vulnerability to adversarial attacks. In this black-box setting, an attacker has access to a single host within a network, with no direct access to the NIDS itself, or the classifiers used. Through experiments on a SYN Flood DDoS

TABLE V
ADVERSARIAL LEARNING - DATA POISONING ATTACKS AND DEFENSES

Research Ref.	Attack(s)	Defense(s)	ML/DL Method(s)	Dataset(s)
Kuppa et al., 2019 [53]	Data poisoning	–	OC-SVM, isolation forests, manifold approximation	CSE-CIC-IDS2018, CICFlowMeter
Paudice et al., 2019 [73]	Label flipping poisoning	Label sanitization	SVM, K-NN	BreastCancer, MNIST, Spambase
Alrawashdeh and Goldsmith, 2020 [14]	Backdoor data poisoning	Adaptive function and pruning	DNN, DBN, CoGAN, L-BFGS & FGSM	ISCX NSL-KDD, Ransomware
Chen et al., 2020 [33]	Label flipping poisoning	Federated learning	GRU-SVM, SGD, DNN	KDD-CUP1999, CIC-IDS2017, WSN-DS
Tolpegin et al., 2020 [101]	Targeted label flipping poisoning	Clustering model gradients	Federated learning, DNN	CIFAR10, Fashion-MNIST
Xu et al., 2020 [113]	Targeted parallel data poisoning	–	LSTM, ConvS2S, CNN	TORCHTEXT (IWSLT2016), News-Commentary v15
Chen et al., 2021 [31]	Data poisoning	Attack-Agnostic Defense, Deep-kNN	GAN, CNN, MSE	MNIST, CIFAR-10, Fourclass, House Pricing
Li et al. 2021 [56]	Backdoor poisoning	–	DNN	Google Play Store, TinyImageNet
Ning et al., 2021 [70]	Clean-label data poisoning backdoor attack	Supervised and unsupervised poison sample detection	AE, NN	MNIST, CIFAR10, ImageNet10, GTSRB
Schwarzschild et al., 2021 [85]	Data poisoning, hidden trigger backdoor	–	SVM, NN, transfer learning	CIFAR10, CIFAR100, TinyImageNet
Severi et al., 2021 [86]	Clean-label backdoor poisoning	Spectral signatures, HDBSCAN, isolation forest	NN, random forest, linear SVM	EMBER, Contagio Malware Dump, DREBIN
Venkatesan et al., 2021 [104]	Backdoor data poisoning	Nested training for sanitization	Linear SVM, NIDS	CyberVAN, MNIST
Wang et al., 2021 [110]	Label flipping poisoning	Stochastic approximation	CNN, RKHS, robust learning	MNIST, CIFAR10
Lai et al., 2023 [54]	Label-flipping, and backdoor attacks	Federated learning (DPA-FL)	IDS, reinforcement learning, DQN, CNN	CIC-IDS2017

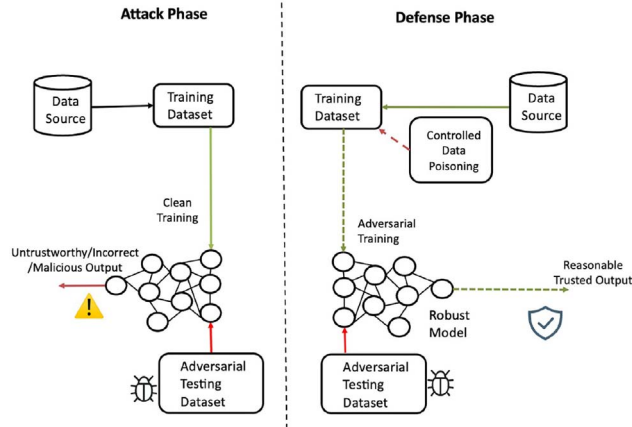


Fig. 6. Test time evasion adversarial learning attacks and defenses for ML/DL-based network intrusion detection systems.

attack scenario, they demonstrated a significant reduction in NIDS detection accuracy using evasion attacks on their SDN NIDS *Neptune*, which used supervised learning on network flow statistics to train and classify live traffic. It was developed with the core goal of detecting DDoS attacks, most notably synchronize (SYN) floods to enable evaluation of adversarial evasion attacks based on attack detection accuracy. Among the classifiers tested, K-nearest neighbors (KNN) proved the most robust, with a single feature perturbation lowering detection accuracy from 100% to 50%. Logistic regression (LR), random forest (RF), and support vector machine (SVM) classifiers were more susceptible to the same perturbations [9].

White Box TTE Attacks: Pioneer work of Biggio et al. introduced a gradient-based evasion technique to deceive SVMs and neural networks classifiers [25]. Goodfellow et al. also made significant contributions to the initial research by utilizing the fast gradient sign method to generate adversarial examples,

successfully causing several classifiers to misclassifying the output. For instance, a shallow softmax classifier was assessed, exhibiting an error rate of 99.9% alongside an average confidence level of 79.3% [39].

Melis et al. were able to show that the robot vision of a humanoid called “iCub” was fooled by adversarial examples crafted from *iCubWorld28* image dataset. They implemented multiclass SVM versions to demonstrate that stealthy adversarial examples forced the model to misclassify [61]. Ayub et al. demonstrated the feasibility of a white-box model evasion attack in intrusion detection, where the attack relies on specific parameters used in the trained model rather than the training dataset. They employed CICIDS 2017 and TRAbID 2017 datasets to implement the multilayer perceptron (MLP) model achieving a baseline accuracy of approximately 99% in classifying attack and benign data. Adversarial samples resulted in a significant decrease in model accuracy, ranging from 20% to

TABLE VI
ADVERSARIAL LEARNING - TEST TIME EVASION ATTACKS AND DEFENSES

Research Ref.	Attack (s)	Defense(s)	ML/DL Method(s)	Dataset(s)
Biggio and Roli, 2018 [26]	Poisoning at inference	Reactive and proactive defenses	ANN, SVM, RF, decision trees	MNIST
Chen et al., 2019 [32]	Transferability, hard-label attacks	Stateful detection defenses	NN, AE, query blinding, FGSM	CIFAR-10, CINIC-10, TinyImageNet
Ayub et al., 2020 [22]	Model evasion	–	MLP, NN, JSMA	CIC-IDS2017, TRAbID2017
Mehanaz et al., 2020 [60]	Model inversion, attribute inference	–	Decision tree, DNN	General society survey, adult census income
Pawlicki et al., 2020 [74]	Misclassification	–	Random Forest, K-NN Classifier, IDS ANN, SVM	CIC-IDS2017
Pierazzi et al., 2020 [76]	Problem-space and feature-space manipulation	–	SVM, greedy algorithm, Stochastic gradient descent	AndroZoo, VirusTotal, DREBIN
Han et al., 2021 [40]	Traffic mutation	Adversarial training, feature reduction	MLP, DT, LR, KitNET, traffic obfuscation, GAN, IF, Lasso Regression	Kitsune, CIC-IDS2017
Talty et al., 2021 [98]	Poisoning at inference	Feature engineering	Random forest, MLP, SVM, logistic regression, KNN classification, SGD, FGSM, PGD	KDD-CUP1999, ISCX, NSL-KDD
Li et al., 2022 [55]	Misclassification	Adversarial training, randomization, projection, detection	DNN, AE, GAN, KDE, LID, ODD, ReBeL, FGSM, Logic Regression	MNIST, CIFAR10
Merzouk et al., 2022 [62]	Feature manipulation	–	FGSM	ISCX, NSL-KDD, UNSW-NB15, CIDD5-001
Zhang et al., 2022 [117]	Poisoning at inference	Model voting, adversarial training, query detection	NIDS, DNN, MLP, CNN, C-LSTM,	CIC-IDS2017, CSE-CIC-IDS2018
Debicha et al., 2023 [36]	Botnet	Adversarial training, anomaly detection	MLP, random forest, K-NN	CTU-13, CSE-CIC-IDS2018
Debicha et al., 2023 [35]	Missclassification	Robust classification and anomaly detection	DNN, FGSM, PGD, DeepFool, Carlini & Wagner, Fusion Rules	ISCX, NSL-KDD, CIC-IDS2017
Hore et al., 2023 [44]	Packet manipulation	–	DT, RF, ML, DNN, SVM, LR	CIC-IDS2017, CSE-CIC-IDS2018
Bostani and Moonsamy, 2024 [27]	Malware	–	SVM, optimization	DREBIN, AndroZoo

30%, for both datasets. As a potential defense strategy, reducing the amplitude of the gradient may enhance the model's generalization ability. By doing so, the model becomes more robust and less susceptible to adversarial attacks, ultimately improving its overall performance and ability to generalize to unseen data [22].

Gray Box TTE Attacks: Biggio et al. employed a gradient-based evasion method demonstrating the vulnerability of these classifiers to adversarial attacks. The effectiveness of attacks against classification algorithms such as SVMs and neural networks revealed that adversarial samples have a high probability of evading detection, even if the adversary only possesses a copy of the classifier learned from a small surrogate dataset. For example, utilizing a small subset (20%) of the PDF corpus samples from the *Contagio* dataset, adversaries were able to craft adversarial examples that successfully evaded the target classifiers' ability to distinguish between legitimate and malicious PDF files [25].

We present the summary of the reviewed work in the area of adversarial TTE, including the attacks, defenses, ML/DL methodologies, and datasets utilized in Table VI.

TTE Defenses: Test time evasion adversarial learning defenses focus on detecting and mitigating adversarial examples that are specifically crafted to deceive the model at the time of testing. By incorporating techniques such as input sanitization, defensive distillation, or ensemble methods, these defenses strive to improve the model's ability to accurately classify inputs even in the presence of adversarial perturbations. Papernot et al. introduced an adversarial defensive technique called defensive distillation. They were able to show empirically that defensive distillation reduced the success rate of adversarial sample crafting from 95.89% to 0.45% against a

deep neural network classifier trained on the MNIST dataset, and from 87.89% to 5.11% against another classifier trained on the CIFAR10 dataset [72].

Pawlicki et al. used test time neuron activations to detect adversarial attacks, employing four evasion attack algorithms: 1) fast gradient sign; 2) basic iterative method; 3) Carlini and Wagner attack; and 4) projected gradient descent. They collected neural activations from an ANN trained on a subset of the CICIDS2017 dataset and adversarial examples. By training and testing five ML classifiers, they achieved a recall of 0.99 for adversarial attacks with RF and nearest neighbour classifier algorithms [74]. Debicha et al. developed a transfer learning-based adversarial detector and evaluated the effectiveness of employing multiple strategically placed detectors in IDS. Through the implementation of state-of-the-art models with several evasion attacks, they demonstrated that combining multiple detectors enhances the detectability of adversarial traffic [35].

As presented in Table VI, most common TTE attack is the use of adversarial samples during inference. This attack manipulates the model's behavior, potentially leading to misclassification. On defense mechanisms, adversarial training is the basic strategy, which involves augmenting the training process by incorporating adversarial examples, which are carefully crafted input samples designed to deceive the model. By exposing the model to these adversarial examples during training, it learns to recognize and appropriately respond to such attacks. Adversarial training improves model generalization, enhances its ability to handle perturbations, and makes it more robust against attacks, reducing vulnerabilities and manipulation of predictions. However, attackers can still adapt and craft sophisticated attacks that can bypass the defenses learned through adversarial training. Therefore, it is crucial to continuously re-

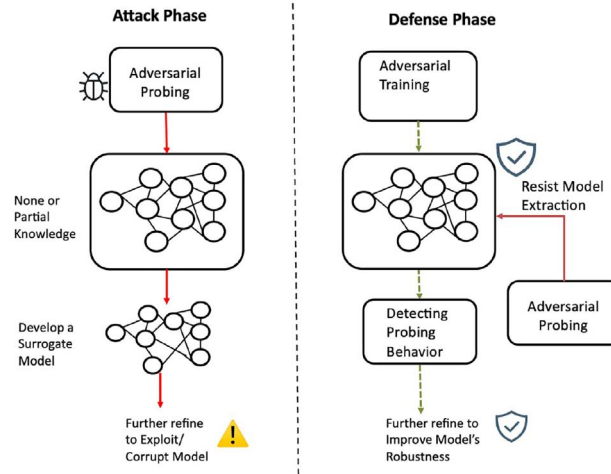


Fig. 7. Reverse engineering adversarial learning attacks and defenses for ML/DL-based network intrusion detection systems.

search and develop new defense strategies to stay ahead of evolving attack techniques. On methodologies applied, researchers applied various ML/DL and preferring neural networks, which is aligned with the overall trend in Adversarial-NIDS field. TTE research is utilizing a wide array of IDS datasets, including established primary datasets such as KDD-CUP1999 and NSL-KDD, alongside contemporary datasets such as UNSW-NB 15, Kitsune, CIC-IDS 2017, CSE-CIC-IDS2018, and ISCX, covering both historical and modern-day attack scenarios.

C. RE Attacks and Defenses

RE attacks exploit the ML pipeline by analyzing feature engineering, extracting sensitive training data, and deducing model architecture, parameters, and internal representations. There are various RE approaches applicable in the field of NIDS. For example, investigating network protocols to understand their structure, behavior, and vulnerabilities aids in effectively detecting and preventing attacks targeting specific protocols. Furthermore, RE techniques assist in malware analysis and signature extraction, enhancing NIDS capabilities in detecting known and unknown threats. Binary code analysis supports vulnerability identification, understanding system responses, and overall security enhancement of NIDS. Fig. 7 shows how adversaries use RE approach for attacks and how the same approach can serve to build a defense strategy against adversarial attacks.

Model inversion attacks involve extracting sensitive information and insights about the training data by reverse-engineering. These attacks pose a significant risk to privacy and require protective measures to safeguard the confidentiality of the model and its data.

A RE attack gains knowledge or extracts information about the underlying detection model or its internal workings to understand and exploit the model's vulnerabilities, weaknesses, or decision-making mechanisms [52]. Nayan et al. addressed the vulnerabilities of on-device model extraction and offered a systematic classification of model extraction attacks and defenses [68]. By analyzing the model's structure, parameters, or outputs, the attacker aims to deduce valuable information,

such as feature representations, detection rules, and sensitive training data. Atwell et al. investigated the threat of reverse TCP attacks to gain remote access to end-user networks by exploiting the connection process [21]. Breier et al. demonstrated the potential of RE in neural networks through fault attacks. By flipping the sign bit, fault attack manipulated intermediate values, thereby enabling the recovery of proprietary model parameters [28]. Attackers leverage RE to exploit vulnerabilities and create potent adversarial examples. On defensive side, RE can serve legitimate purposes, such as model interpretation, debugging, auditing, accountability, and transparency [46].

Black Box RE Attacks: By exploring model inversion attacks, Mehnaz et al. investigated adversaries who can use known nonsensitive attributes to infer sensitive attributes. Two new attack methods, confidence modeling-based and confidence score-based, are introduced. Decision tree and deep neural network models trained on real datasets are evaluated for such attacks. Vulnerability to model inversion attacks is identified within specific attribute-based groups, such as gender or race [60].

Gray Box RE Attacks: In a gray-box backdoor attack called *DeepPayload*, Li et al. introduced a practical RE attack scenario where the attacker has access to the compiled DNN model in the app, and does not have access to the original training data or metadata used for training. Using RE, a decompiled DNN model into a data-flow graph, an attacker can inject a payload consisting of a resize operator, trigger detector, and output selector. The modified model, incorporating the injected components, altered its behavior based on the trigger presence probability, allowing for the attacker's desired output. This attack uses bytecode reverse-engineering by directly manipulating the dataflow graph to inject malicious logic directly into deployed DNN model. The study reveals that this attack effectively triggers the malicious payload with a high success rate (93.5%), with minimal latency overhead (2 ms) and accuracy decrease (1.4%). This RE attack focuses on distributed or deployed models beyond developers' control. Unfortunately,

existing defense techniques require training or testing with extensive sample sets, making them unsuitable for deployed models [56].

RE Defenses: RE was employed by Wang et al. to detect and prevent backdoor attacks on deep neural networks. They identified hidden triggers within DNNs and developed three defense methods: 1) an early filter for adversarial inputs; 2) neuron pruning-based model patching; and 3) unlearning. The detection of backdoor injection is based on anomaly measurement of infected and clean model by how much the label with smallest trigger deviates from the remaining labels. To mitigate these attacks, they analyzed three methods, including an early filter for adversarial inputs, a model patching algorithm based on neuron pruning, and another based on unlearning. By pruning 30% of neurons reduces attack success rate to nearly 0% with the slight reduction of classification accuracy by 5.06%. Training the DNN model with reverse engineered triggers proved effective in unlearning the original trigger, resulting in an attack success rate of less than 6.70% and a maximum reduction in classification accuracy of 3.6% [106].

Asnani et al. presented threat of model parsing for generative models (GMs). The objective is to infer network architectures and training loss functions from generated images to address concerns related to the misuse of GMs. Their proposed framework consists of two networks: 1) the fingerprint estimation network (FEN); and 2) the parsing network (PN). These networks estimate fingerprints from images and predict the corresponding model parameters. A dataset of 100 000 images from 116 GMs was collected to demonstrate parsing of the hyperparameters [19].

Table VII presents an overview of studied RE ML/DL methods applied for adversarial attacks and defenses. As we observe that standard definition of protocols and identification and malware analysis can be used as defense mechanisms for protocol manipulation and malware attacks. Model and data information can be reverse engineered through model extraction, model inversion, and model parsing by using deep neural networks and generative adversarial networks.

There are security concerns associated with RE the architecture of a DNN model by exploiting the memory access pattern of a processor during DNN execution. To counter such attacks, Liu et al. proposed a defensive mechanism that combines techniques such as oblivious shuffle, address space layout randomization (ASLR), and dummy memory accesses. This defense aimed to obfuscate the memory access pattern and minimize the risk of such attacks with minimal overhead. The effectiveness of this proposed defense was evaluated through a modified attack on an existing model. The results illustrated that the incorporation of these techniques notably heightened the complexity of the attack while keeping memory access overhead low. Moreover, the defense mechanism was scalable, made it applicable to larger DNN models by effectively mitigating RE attacks that might exploit memory access patterns [58]. The defense strategy presented by Xiang et al. introduces an optimization-based approach to counter RE, applicable both before and during classifier training. This defense primarily targets backdoor attacks by estimating the recognized pattern

employed in the attack. Upon identifying backdoor poisoning and the target class, the estimated backdoor pattern is used to pinpoint and eliminate the poisoned data directly from the training set [112].

VI. LIMITATIONS AND CHALLENGES

Our objective is to provide a comprehensive and active reference in the field of adversarial learning attacks and defenses for network intrusion detection systems; however, there are challenges and limitations within our work. First, the availability of comprehensive literature covering all relevant aspects of adversarial learning is limited, hindering direct comparisons and in-depth analysis of research breadth of the topic. While there have been significant advancements in the broader field of adversarial learning attacks and defenses, the specific subareas of deep learning, network intrusion, DP, TTE, and RE have seen relatively fewer studies. Additionally, our focused approach on a shorter time frame may result in a narrower scope of research discussed. Furthermore, the dynamic and rapidly evolving nature of the field presents challenges in effectively capturing the latest advancements in adversarial learning, emphasizing the need for future studies to stay current.

VII. DISCUSSION

Research in adversarial learning within the network domain continues to evolve, focusing on advancing attack detection capabilities and developing more robust defense strategies. The relevance of existing older yet popular NIDS datasets for advanced ML/DL algorithms remains a concern. ML/DL-generated synthetic data may address challenges of data scarcity, scalability, and attack rarity effectively. Majority of existing research studies highly rely on outdated data sources. These outdated data sources suffer from a high percentage of duplicate records and highly imbalanced attack categories. Accurate data interpretation is crucial in ML-based NIDS for representing normal and attack scenarios. While diverse attack datasets help identify and mitigate threats, adversaries exploit this knowledge to craft deceptive attacks. Challenges such as limited availability, imbalanced class distribution, evolving techniques, and privacy concerns hinder the effectiveness of network attack datasets. Keeping datasets up-to-date becomes a challenge due to evolving techniques and the need for real attack instances [15], [89], [100]. To compare the scope, applicability, and limitations of these datasets, Ghurab et al. analyzed the benchmark datasets (KDD99, NSL-KDD, Kyoto 2006+, ISCX 2012, UNSW-NB15, CIDD5-001, CICIDS2017, CSE-CIC-IDS2018), and recommended to utilize more recent datasets representing modern-day attacks [38]. However, another study by Alshamy et al. presented that 66% of studied 39 NIDS ML-DL classification research works (2017-2020) utilized KDD99/NSL-KDD datasets with 46% using KDD99 dataset [15]. Similarly, an evaluation of ML-DL performance study by Pektas and Acarman showed that 80% of 10 studies (2008-2018) utilized KDD99/NSL-KDD datasets, while a 2020 review by Thakkar et al. presented that all 6 studied research work for ML-DL NIDS utilized NSL-KDD datasets [75], [100].

TABLE VII
ADVERSARIAL LEARNING - REVERSE ENGINEERING ATTACKS AND DEFENSES

Research Ref.	Attack(s)	Defense(s)	ML/DL Method(s)	Dataset(s)
Antunes et al., 2011 [17]	Protocol Manipulation	Standard Protocol Identification	Moore Reduction	CSIC 2010, Salient Person, FTCDATA
Alabdulmohsin et al., 2014 [10]	Spam	Randomized diversification	Adversarial Learning, Linear SVM	UCI datasets
Atwell et al., 2016 [21]	Reverse TCP Attacks, Social Engineering, Zero Day	Security Policy, Security Awareness, Incident Response	TCP analysis	Click (Port 80 HTTP), VirusTotal
Tramer et al., 2016 [102]	Model Extraction, Model Inversion	–	NN, Decision Trees, SVM	Digits, Adult, Statlog (German Credit Data), Steak Survey
Papernot et al., 2017 [71]	Misclassification	Reactive and Proactive - Gradient Masking	DNN	MNIST, GTSRB
Liu et al., 2019 [57]	Model Extraction and Inversion	Address Space Layout Randomization (ASLR), DumMA	DNN, CNN	Memory Access Patterns
Wang et al., 2019 [108]	Model Manipulation	L-AWA-maxKL	DNN	MNIST
Wang et al., 2019 [106]	Backdoor Attacks - BadNets & Trojan	Filtering, Neuron Pruning	DNN	MNIST, GTSRB, Youtube Face, PubFig
Breier et al., 2021 [28]	Fault Attacks	–	NN	TinyImageNet, CIFAR-10
Li et al., 2021 [56]	Injected Backdoor Attack	–	DNN	Google Play Store
Ismael and Thanoon, 2022 [46]	Malware	Standard malware identification	Malware Analysis	CIC-MalMem-2022
Asnani et al., 2023 [19]	Model Parsing	Fingerprint Estimation Network (FEN), Parsing Network (PN)	Generative Models (GM), CNN, GAN, VAE	MNIST, GTSRB, Tiny ImageNet, CelebA, FaceForensics++

Ring et al. compared 34 datasets, representing research works on dataset generation or utilization, focusing on features per dataset [79]. For a thorough evaluation of ML/DL classifiers, Sarhen et al. introduced a standardized NetFlow-based feature set consisting of 43 features. They tailored NIDS datasets (UNSW-NB15, BoT-IoT, ToN-IoT, CSE-CIC-IDS2018) using the define standard NetFlow-based features to enhance the consistency in comparing the performance evaluation [1], [83]. The growing importance of deep learning in enhancing predictive modeling and decision-making makes them attractive for NIDS. However, adversarial research specific to ML/DL-based NIDS remains limited compared with non-NIDS domains. The challenges of adapting image-specific learning algorithms to NIDS have driven the exploration of domain-specific methods [41], [51], [80].

VIII. CONCLUSION AND FUTURE DIRECTIONS

Machine learning and deep learning models excel at adapting to evolving threats and detecting subtle anomalies, making them invaluable for dynamic cybersecurity environments. However, their reliance on substantial computational resources can hinder real-time detection in high-speed networks. Limited interpretability further challenges understanding their decisions, emphasizing the need for current, realistic network datasets to detect adversarial attacks and develop robust defenses. Future research on optimizing ML/DL models through lightweight architectures, incorporating explainability techniques, and integrating traditional approaches such as rule-based systems and ensemble methods can enhance NIDS performance and reliability. The limited interpretability of evolving machine learning methods in NIDS hinders intrusion validation and resource optimization in security-focused systems.

Adversarial learning in NIDS presents critical challenges and opportunities for future research directions.

- 1) *Dataset Quality and Realism*: Develop authentic datasets reflecting current attack scenarios and employ ML/DL-generated synthetic data and data augmentation techniques to tackle data scarcity and scalability challenges effectively.
- 2) *Interpretability*: Employ explainable AI techniques to validate detected intrusions, support expert analysis, and optimize resource allocation.
- 3) *Domain-Specific Algorithms*: Develop adversarial learning algorithms tailored to the unique characteristics of network data.
- 4) *Real-World Evaluation*: Bridge the gap between theory and practice by testing real-world environments to ensure usability and effectiveness.
- 5) *Zero-Day Threat Detection*: Explore novel techniques and threat intelligence to handle to zero-day threats effectively.

Achieving robust generalization and accurate predictions for adversarial threats remains a critical challenge in NIDS domain. There is a significant gap in NIDS-specific DP, TTE, and RE. Our work emphasizes the need for continuous evaluation and advancement of adversarial learning techniques to ensure defenses remain effective against evolving adversarial threats in network intrusion domain.

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